

EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

CONSUMER PRICE TREND ANALYTICS

Aim

To import, transform, analyze, and visualize the All India Consumer Price Index (CPI) dataset using Power BI Desktop and create interactive analytical views and a dashboard for understanding consumer price trends, sector-wise differences, and category-level patterns.

Dataset: All India Consumer Price Index (2013–2023)

Dataset size: 365 records × 30 attributes

Project focus: Consumer Price Trend Analytics

Primary dimensions: Sector, Year, Month

Primary measure: General index

Key analytical categories: Food and beverages, Clothing, Housing, Fuel and light, Health, Transport and communication, Education and Miscellaneous

Algorithm 1

Importing CPI Data into Power BI

- Step 1: Open Power BI Desktop.
- Step 2: Select Home → Get Data.
- Step 3: Choose Text/CSV.
- Step 4: Browse and select All India Consumer Price Index.csv.
- Step 5: Select Transform Data to open Power Query before loading.

Algorithm 2

Data Transformation using Power Query

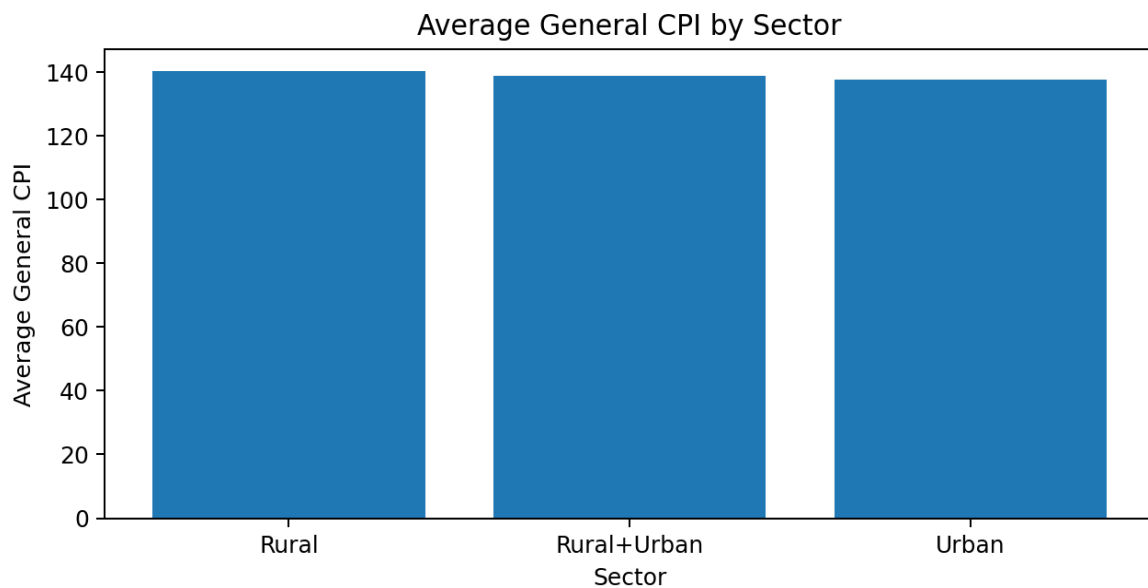
- Step 1: Open Power Query Editor.
- Step 2: Remove duplicate rows.
- Step 3: Convert CPI fields to numeric data types.
- Step 4: Handle null values, especially missing Housing values.
- Step 5: Standardize Sector, Year and Month fields.
- Step 6: Create a Date field from Month and Year.
- Step 7: Apply the transformations and load the cleaned table.

Procedure — Data Import and Transformation

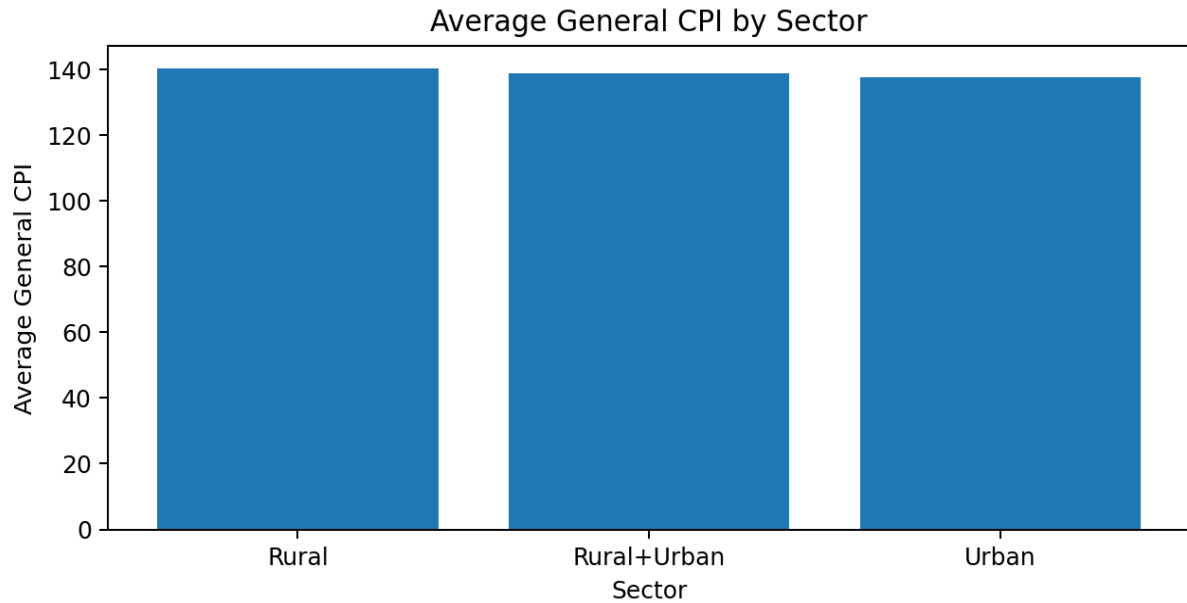
1. The Power BI data preview window displays the imported CPI table. Select Transform Data instead of Load to open Power Query.
2. Examine the table and verify the important fields: Sector, Year, Month, Food and beverages, Clothing, Housing, Fuel and light, Health, Transport and communication, Education, and General index.
3. Remove completely blank rows using Home → Remove Rows → Remove Blank Rows.
4. Select the table columns and use Home → Remove Rows → Remove Duplicates.
5. Standardize the Sector values so that Rural, Urban and Rural+Urban are represented consistently.
6. Set Year as Whole Number and Month as Text.
7. Convert CPI category columns and General index to Decimal Number.
8. Handle missing Housing values by replacing them with an appropriate value such as the column mean or a Power Query replacement rule.
9. Verify that no invalid text values remain in the numerical CPI fields.
10. Create a derived Date column by combining Month and Year. This enables time-series analysis.
11. Review the Applied Steps pane and verify the transformation sequence.
12. Select Home → Close & Apply to load the cleaned data into the Power BI model.

Procedure — CPI Model Preparation

13. Switch to Data/Table view and inspect the transformed CPI table.
14. Confirm that duplicate records and blank rows have been handled.
15. Confirm that the numerical CPI fields are correctly typed and that the General index is available for analysis.
16. Create a measure named Average General CPI using the AVERAGE function over the General index field.
17. Create Minimum CPI and Maximum CPI measures using MIN and MAX.
18. Create a Latest CPI measure or use the most recent Date with the General index to represent the latest available CPI value.
19. Create a clustered column chart with Sector on the X-axis and Average General CPI as the Y-axis.
20. Verify that the visual compares Rural, Urban and Rural+Urban sectors.



Output — Sector-wise CPI Comparison



Interpretation: The Rural, Urban and Rural+Urban sectors show different average General CPI levels. The comparison makes sector-level price differences immediately visible.

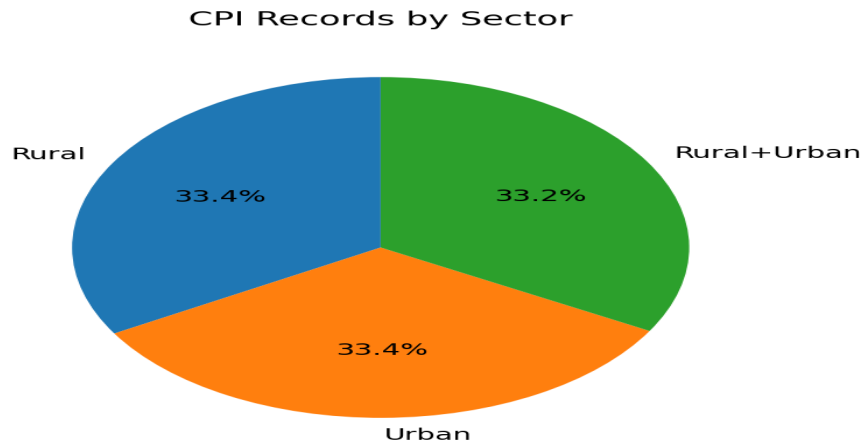
Algorithm 3

Creating a Bar Chart

- Step 1: Choose Clustered Bar/Column Chart.
- Step 2: Drag CPI category to the category axis.
- Step 3: Drag Average General CPI to the value axis.
- Step 4: Sort the visual in descending order.
- Step 5: Add a descriptive title and data labels.

Algorithm 4 — Creating a Pie Chart

- Step 1: Select Pie Chart visualization.
- Step 2: Drag Sector to Legend.
- Step 3: Drag a record-count measure to Values.
- Step 4: Enable percentage labels.
- Step 5: Add the title 'CPI Records by Sector'.



Interpretation: The dataset is organized across Rural, Urban and Rural+Urban observations. The pie chart communicates the composition of the available CPI records.

Algorithm 5 — Creating KPI Cards

Create the following measures in Power BI using the CPI table:

Measure	DAX / Definition	Purpose
Average General CPI	AVERAGE('CPI'[General index])	Overall average CPI
Minimum CPI	MIN('CPI'[General index])	Lowest observed CPI
Maximum CPI	MAX('CPI'[General index])	Highest observed CPI
Latest CPI	Value for latest Date	Most recent CPI observation

Average General CPI	138.81
Minimum CPI	104.00
Maximum CPI	178.00
Latest Rural+Urban CPI	177.20

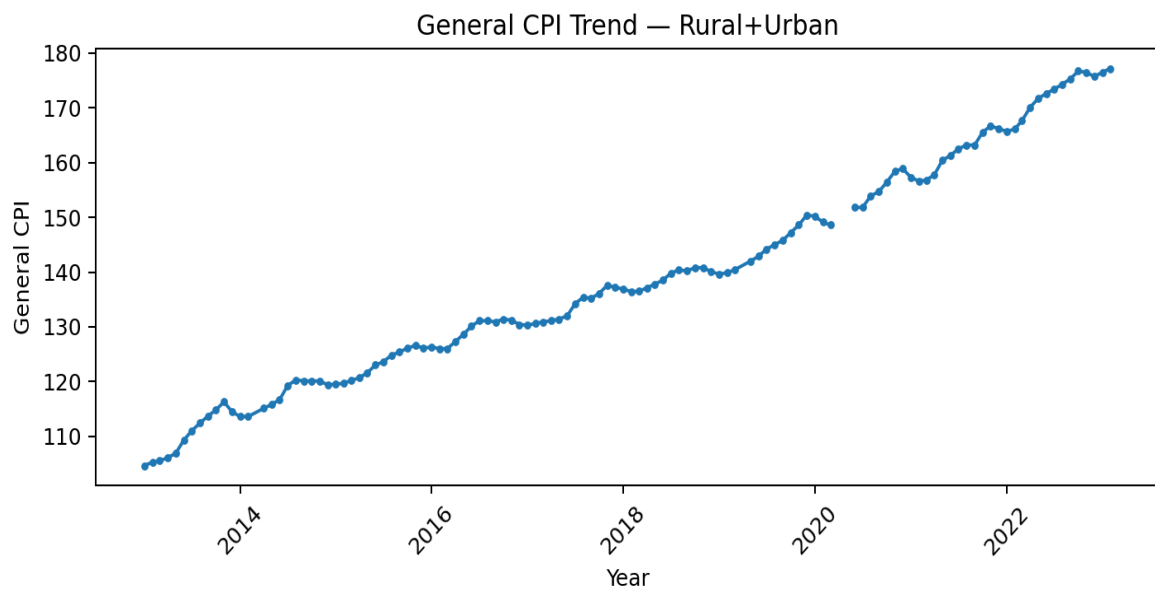
Output values are calculated from the supplied CPI dataset. Latest Rural+Urban observation date: February 2023.

Algorithm 6 — Creating Slicers

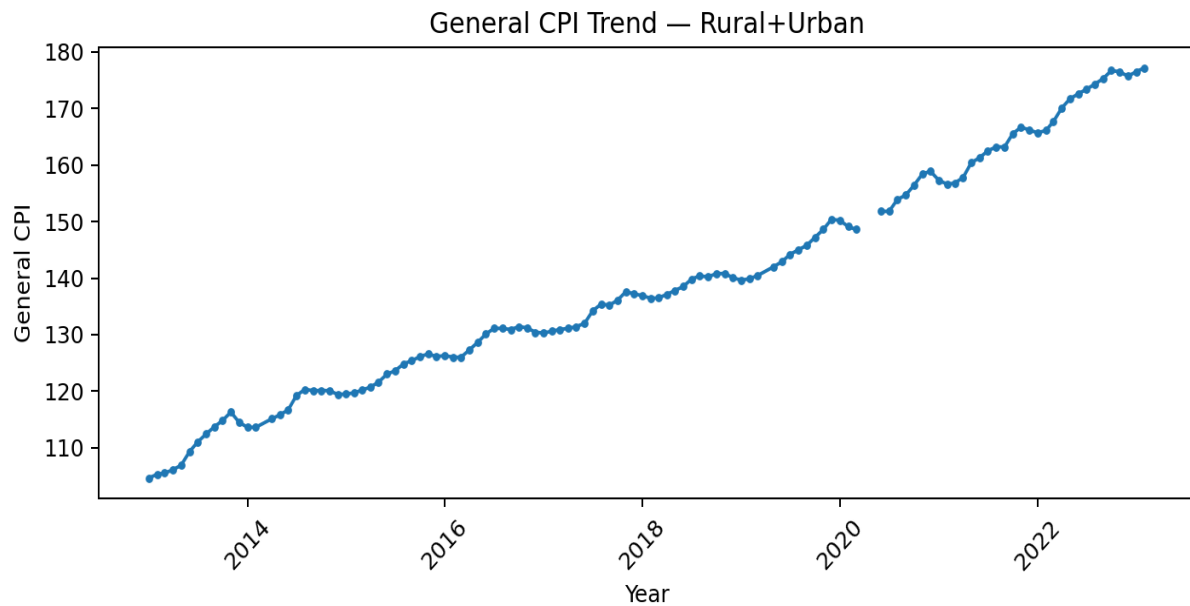
- Step 1: Select the Slicer visualization.
- Step 2: Drag Sector into the slicer field.
- Step 3: Enable single selection when a focused comparison is required.
- Step 4: Apply Rural, Urban, or Rural+Urban filters.
- Step 5: Observe the KPI cards and charts update according to the selected sector.

Example filtered analysis

Filter	Average General CPI	Purpose
All sectors	138.81	Overall CPI view
Rural	140.18	Rural price level
Urban	137.59	Urban price level
Rural+Urban	138.66	Combined CPI view

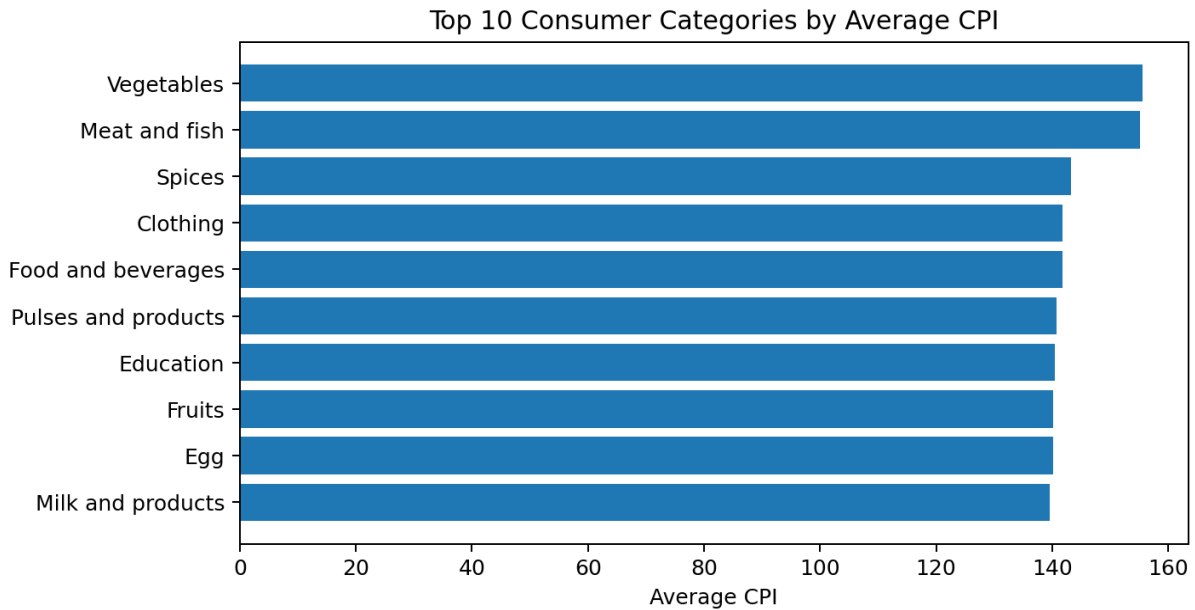


Time-Series Visualization — General CPI Trend



Interpretation: The line chart shows the movement of the General CPI over time for the Rural+Urban sector. This is a core data-visualization use case because a temporal pattern is easier to identify from a continuous line than from raw tabular values.

Category Analysis — Bar Chart

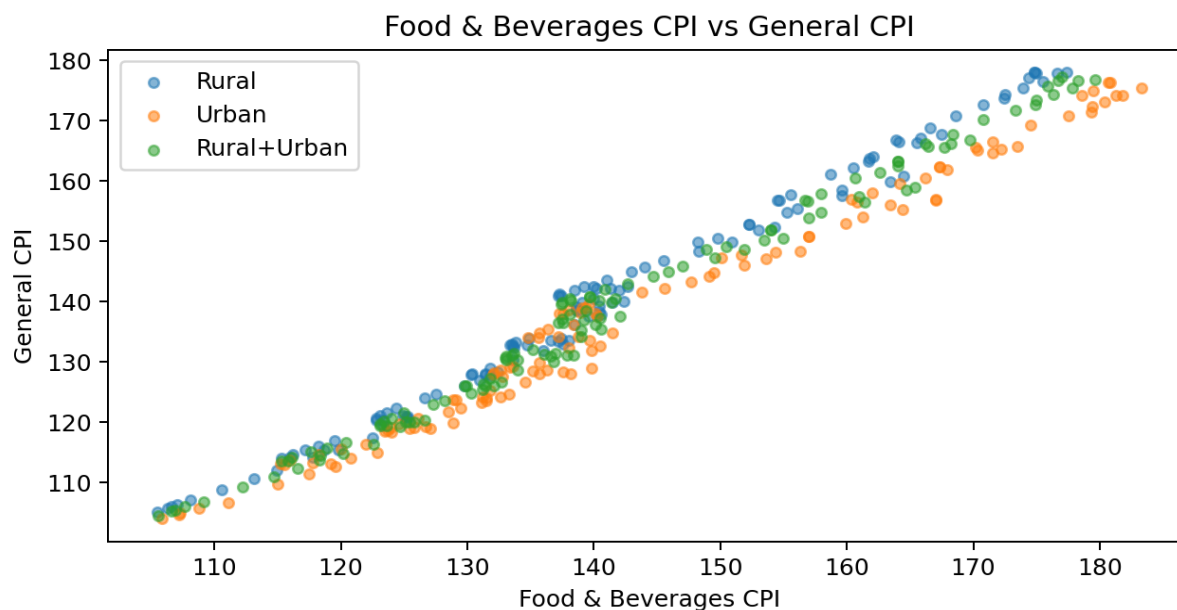


Interpretation: The bar chart ranks consumer categories according to their average CPI values. Sorting categories makes it easier to identify the categories with comparatively higher index levels.

Algorithm 7 — Creating a Trend and Category Visual

- Use Date on the X-axis and General index on the Y-axis for a line chart.
- Use a consumer category field on the category axis and Average CPI on the value axis for a ranked bar chart.
- Apply data labels and meaningful titles.
- Use the Sector slicer to make the analysis interactive.

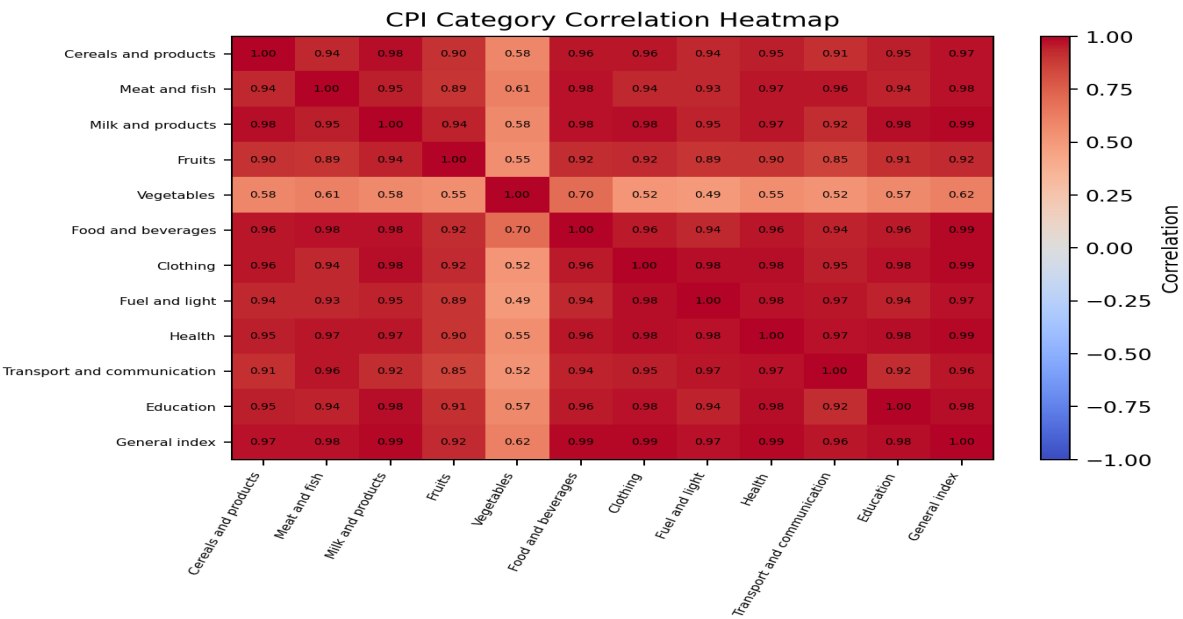
Exploratory Visual — Food & Beverages vs General CPI



Interpretation: The scatter plot examines the relationship between Food and Beverages CPI and the General CPI. Sector labels allow the same relationship to be viewed across Rural, Urban and Rural+Urban observations.

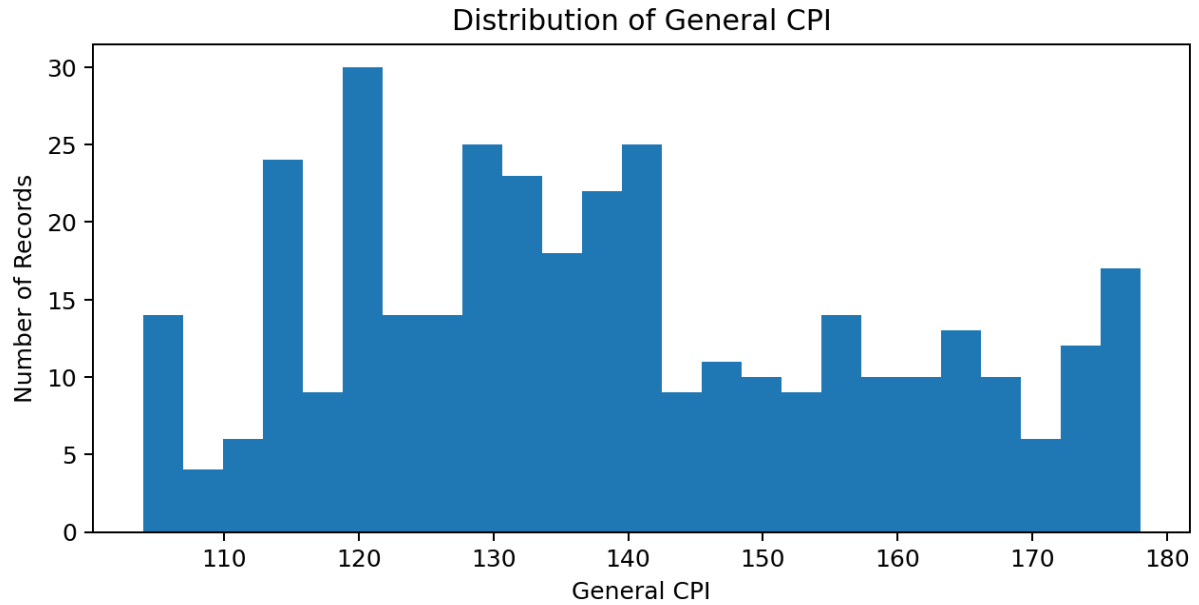
Additional Power BI visual: A scatter chart can use Food and beverages as the X-axis, General index as the Y-axis, Sector as Legend, and Date or another measure as a tooltip.

Correlation Analysis



Interpretation: The correlation heatmap summarizes the strength and direction of relationships among selected CPI categories. Values closer to +1 indicate stronger positive association, while values closer to -1 indicate stronger negative association.

Distribution Analysis



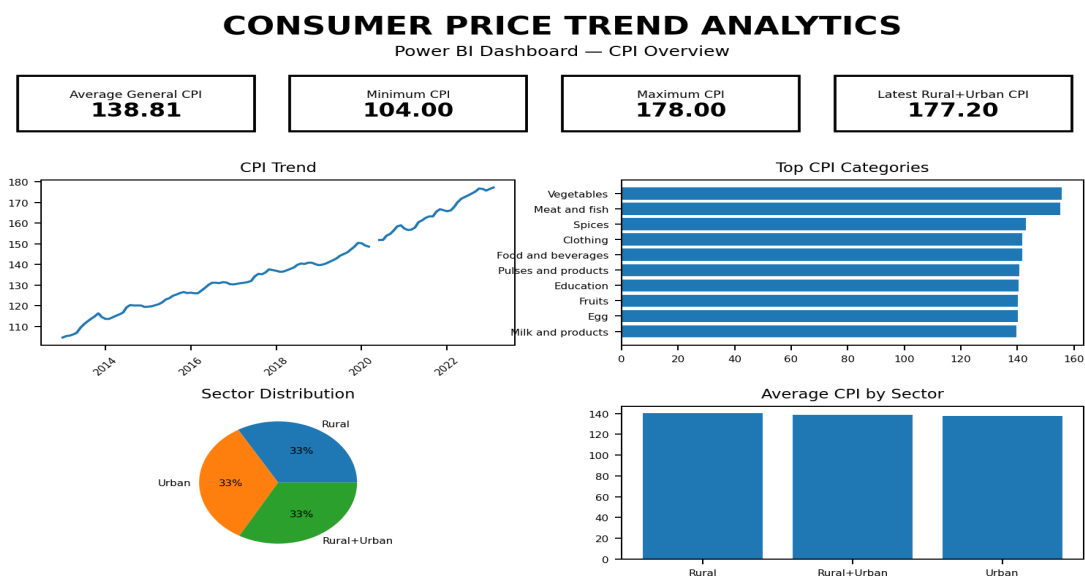
Interpretation: The histogram shows how General CPI observations are distributed. It helps identify concentration, spread and unusual values that may require further investigation.

Box Plot in Power BI: A box-and-whisker visual can be used to examine CPI spread and potential outliers across sectors or categories.

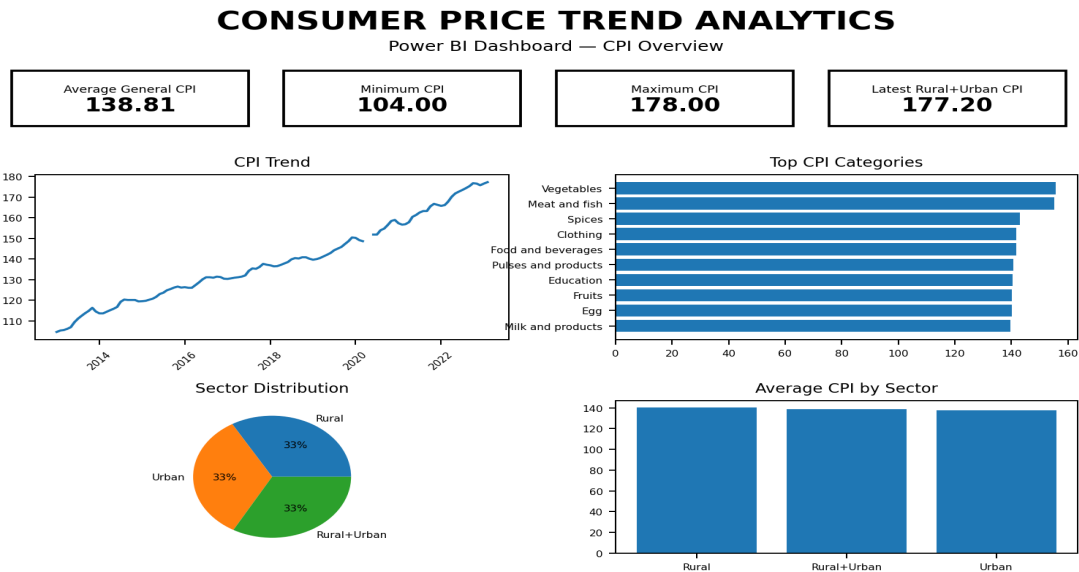
Pie Chart: The sector distribution visual complements the numerical analysis by showing the composition of observations.

Algorithm 8 — Creating the CPI Dashboard

- Step 1: Add KPI Cards for Average, Minimum, Maximum and Latest CPI.
- Step 2: Add a line chart for the General CPI trend.
- Step 3: Add a ranked bar chart for major CPI categories.
- Step 4: Add a sector distribution pie chart.
- Step 5: Add a Sector slicer for interactive filtering.
- Step 6: Arrange the visuals into a single report page.
- Step 7: Add descriptive titles, labels and tooltips.
- Step 8: Save the Power BI report.



Output and Result



Result: The Consumer Price Index dataset was successfully imported, transformed, analyzed, and visualized using a Power BI-style workflow. The resulting dashboard combines KPI cards, sector comparisons, category rankings, time-series analysis, and interactive filtering to communicate CPI patterns effectively.

How the project aligns with Data Visualization:

- Comparison — bar charts compare average CPI across sectors and categories.
- Trend analysis — line charts reveal changes in CPI over time.
- Composition — pie charts communicate sector-wise record distribution.
- Distribution — histograms and box plots show CPI spread and potential outliers.
- Relationship analysis — scatter plots examine associations between CPI variables.
- Correlation — heatmaps summarize relationships among multiple consumer categories.
- Interactive analytics — slicers and KPI cards allow focused exploration.
- Dashboard design — multiple complementary visuals are combined into one decision-support view.

Conclusion: The experiment demonstrates that Power BI can transform the CPI dataset from a raw table into an interactive visual analytics solution. The dashboard supports quick comparison, trend identification, category analysis, and sector-level exploration of consumer prices.

Dataset: All India Consumer Price Index (2013–2023)

Project: Consumer Price Trend Analytics